

University of Eldoret
Comp 314 Advanced Database Design CAT [30 Marks]

1. Define the term normalization with respect to logical design. [1 Marks]
2. Outline two characteristics of a well-designed database. [1 Marks]
3. Explain data fragmentation as used in distributed database systems. (2 Marks)
4. Explain two reasons for one using an online database. [2 Marks]
5. Discuss any two problems associated with concurrency access to databases. [2 Marks]
6. illustrate functional dependency as used in database relation. (2 Marks)
7. Outline two characteristics of each of the following distributed.
 - i. Heterogeneous [2 Marks]
8. Discuss three anomalies that are associated with bad database design. [6 Marks]
9. Explain the query mechanism as used in database management system. [4 Marks]
10. Explain any two database security threats in the context of security analysis and management. [4 Marks]
11. Describe each of the following components of the database tier architecture. [4 Marks]
 - i. Application tier,
 - ii. Presentation tier,